



APPROACH TO WHEEL-RAIL PROBLEMS SOLUTION ON METROS AND LIGHT RAIL SYSTEMS

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Italy / AB Consulting**

B O S T O N



ANDREA BRACCIALI

Bio notes

M.Sc. Mechanical Engineer in 1988

Full professor since 2004 (Rolling Stock Design and Manufacturing, Integrated Design of Railway Systems)

Active on W/R issues on both rolling stock and track issues

Organizer of CM2009 in Florence

Founder of AB Consulting (1994)

Patents on wheels, wheelsets, bogies

Forensic activities on main railway accidents in the last 20 years

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www.andreabracciali.it

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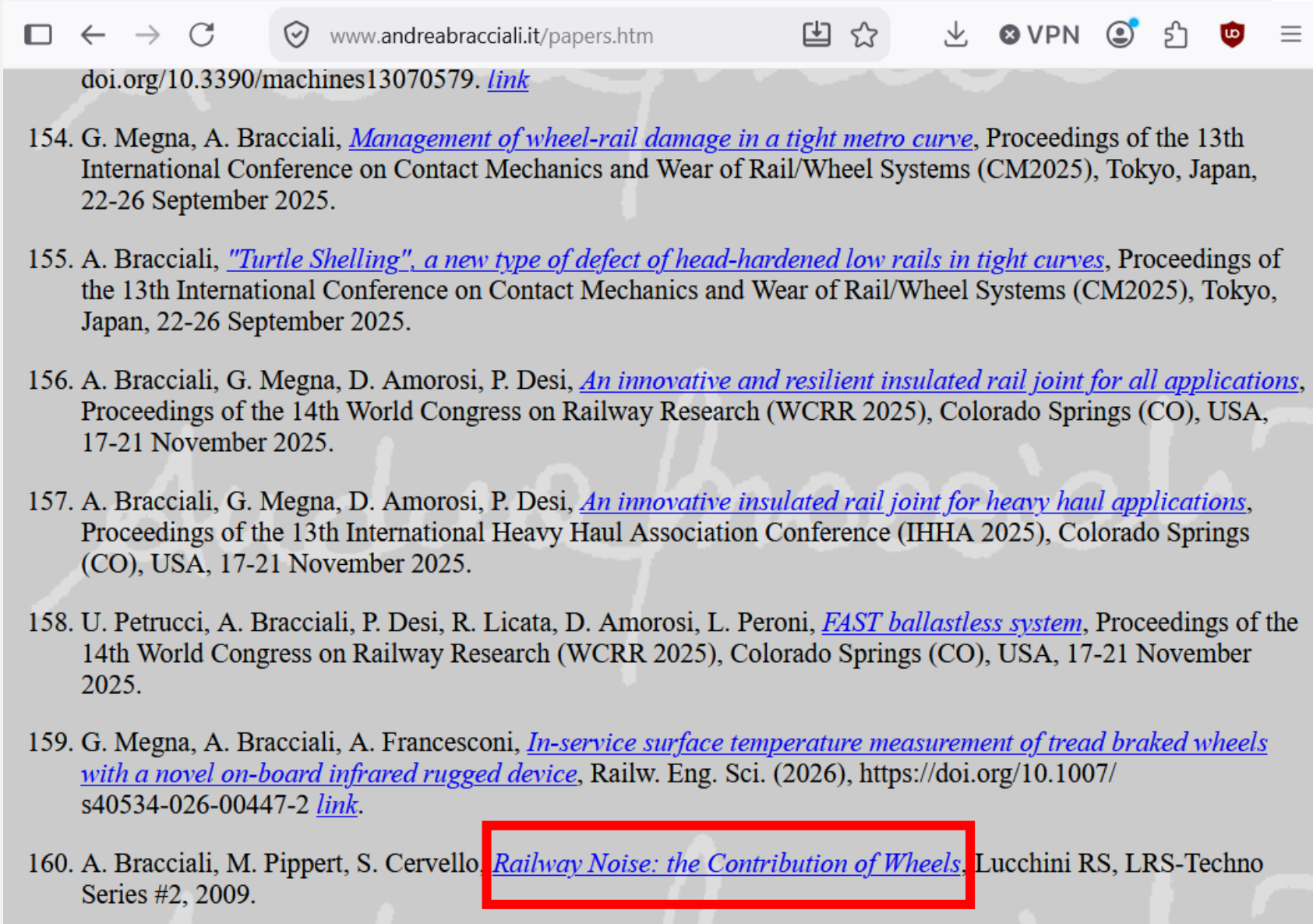


Photo: Giulio Bracciali

Prof. Andrea Bracciali

Welcome!!!

[List of downloadable publications](#) / [Proceedings of International and Italian Conferences](#)



doi.org/10.3390/machines13070579. [link](#)

154. G. Megna, A. Bracciali, [Management of wheel-rail damage in a tight metro curve](#), Proceedings of the 13th International Conference on Contact Mechanics and Wear of Rail/Wheel Systems (CM2025), Tokyo, Japan, 22-26 September 2025.

155. A. Bracciali, ["Turtle Shelling", a new type of defect of head-hardened low rails in tight curves](#), Proceedings of the 13th International Conference on Contact Mechanics and Wear of Rail/Wheel Systems (CM2025), Tokyo, Japan, 22-26 September 2025.

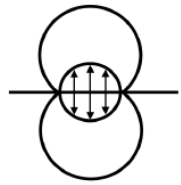
156. A. Bracciali, G. Megna, D. Amorosi, P. Desi, [An innovative and resilient insulated rail joint for all applications](#), Proceedings of the 14th World Congress on Railway Research (WCRR 2025), Colorado Springs (CO), USA, 17-21 November 2025.

157. A. Bracciali, G. Megna, D. Amorosi, P. Desi, [An innovative insulated rail joint for heavy haul applications](#), Proceedings of the 13th International Heavy Haul Association Conference (IHHA 2025), Colorado Springs (CO), USA, 17-21 November 2025.

158. U. Petrucci, A. Bracciali, P. Desi, R. Licata, D. Amorosi, L. Peroni, [FAST ballastless system](#), Proceedings of the 14th World Congress on Railway Research (WCRR 2025), Colorado Springs (CO), USA, 17-21 November 2025.

159. G. Megna, A. Bracciali, A. Francesconi, [In-service surface temperature measurement of tread braked wheels with a novel on-board infrared rugged device](#), Railw. Eng. Sci. (2026), <https://doi.org/10.1007/s40534-026-00447-2> [link](#).

160. A. Bracciali, M. Pippert, S. Cervello, [Railway Noise: the Contribution of Wheels](#), Lucchini RS, LRS-Techno Series #2, 2009.



11th INTERNATIONAL CONFERENCE ON CONTACT MECHANICS AND WEAR
OF RAIL/WHEEL SYSTEMS (CM2018)

122 DELFT, THE NETHERLANDS, SEPTEMBER 24-27, 2018

TRACTION AND CURVING OF VEHICLES EQUIPPED WITH GUIDED INDEPENDENTLY ROTATING WHEELS

Andrea Bracciali, Gianluca Megna*

*7th International Conference on Contact Mechanics and Wear of Rail/Wheel Systems
(CM2006), Brisbane, Australia, September 24-26, 2006*

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RAIL CORRUGATION GROWTH IN A METRO CURVE

Andrea Bracciali

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GROUNDBORNE VIBRATIONS FROM UNDERGROUND METROS: THE EXPERIENCE IN METRONAPOLI

Prof. A. Bracciali¹, Dr. F. Piccioli¹, Dr. V. Orazzo²



ADVANCED FEM ANALYSIS OF SUPPORT BEAM OF A MODERN TRAM

81

Prof. A. Bracciali, Dr. F. Piccioli, T. De Cicco

International Journal of Railway Technology
Volume 1, Issue 1, Pages 231-247, 2012.
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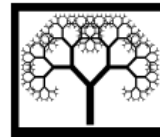
Wayside Train Monitoring Systems: A State-of-the-Art and Running Safety Implications

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Università degli Studi di Firenze, Italy

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Paper 117



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Proceedings of the Second International Conference on
Railway Technology: Research, Development and
Maintenance, J. Pombo, (Editor),
Civil-Comp Press, Stirlingshire, Scotland.

Restoration of Switch Manganese Steels Crossings by Electric Arc Welding in a Robotized Plant

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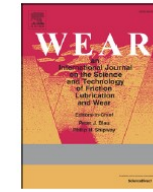


Contents lists available at [ScienceDirect](https://www.sciencedirect.com)

Wear

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Journal homepage: www.elsevier.com/locate/wear



Contact mechanics issues of a vehicle equipped with partially independently rotating wheelsets

Andrea Bracciali, Gianluca Megna*

University of Florence, Department of Industrial Engineering, via di Santa Marta 3, 50139 Florence, Italy



Running dynamics and contact mechanics comparison of two freight bogies running in plain line and through switches and crossings

129

G. Megna¹, H. Magalhaes^{2,3}[0000-0001-7070-7958], Y. Bezin²[0000-0002-0599-8696] and A. Bracciali¹

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² Institute of Railway Research, University of Huddersfield, Huddersfield, UK

³ LAETA, IDMEC, Instituto Superior Técnico, Universidade de Lisboa, Portugal

Minimization of Impact Forces on Crossings by Railhead Profile Optimization

122

G. Megna and A. Bracciali

**Department of Industrial Engineering of Florence (DIEF),
University of Florence
Italy**





A. BRACCIALI - M. PIPPERT - S. CERVELLO

Railway noise: the contribution of wheels

BASICS, THE LEGAL FRAME, LUCCHINI RS PRODUCTS

Series

LRS-TECHNO

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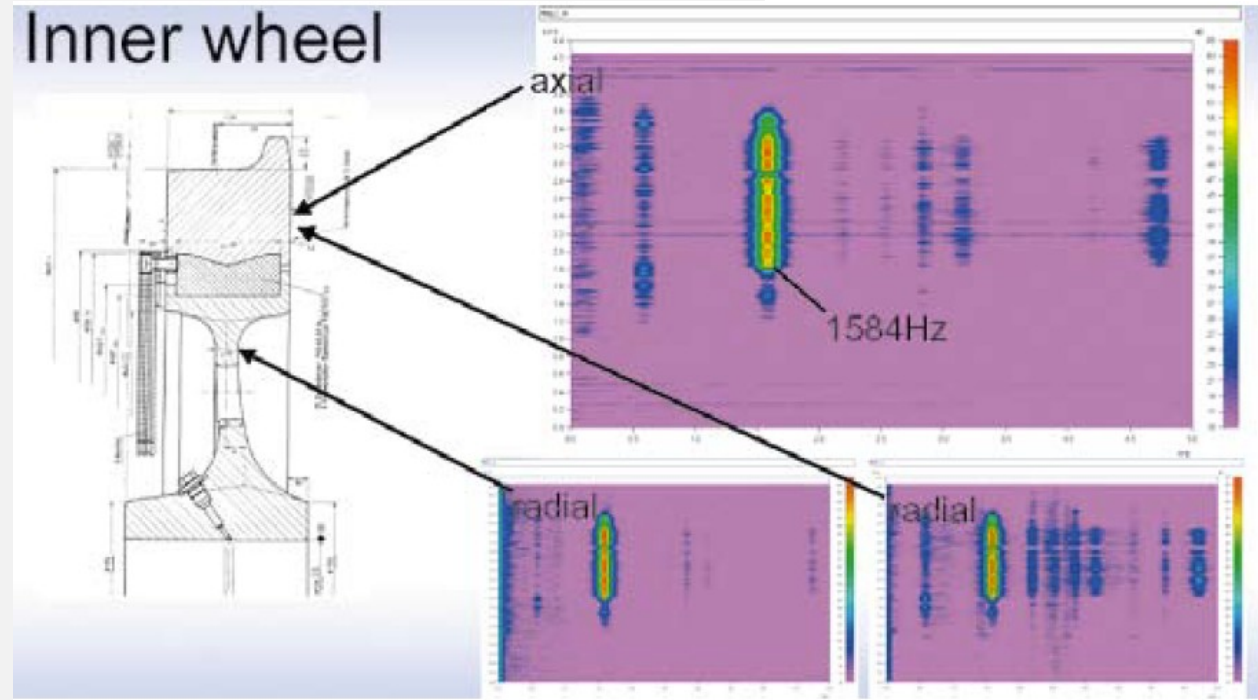


Figure 2.7 Squeal noise from a composite tram wheel. Despite the use of rubber elements between the web and the tyre and the application of an external damper, squeal is present at extremely high levels [5].



Where mass transit systems and railways differ

- Isolated
- Always the same vehicles (at the same speed?)
- Tunnels
- Very sharp curves
- PTA







(Snakes may be dangerous, don't you know?)





Nightmares in the dead of the night



WHEEL-RAIL ISSUES (or why we are here)



RAIL WEAR

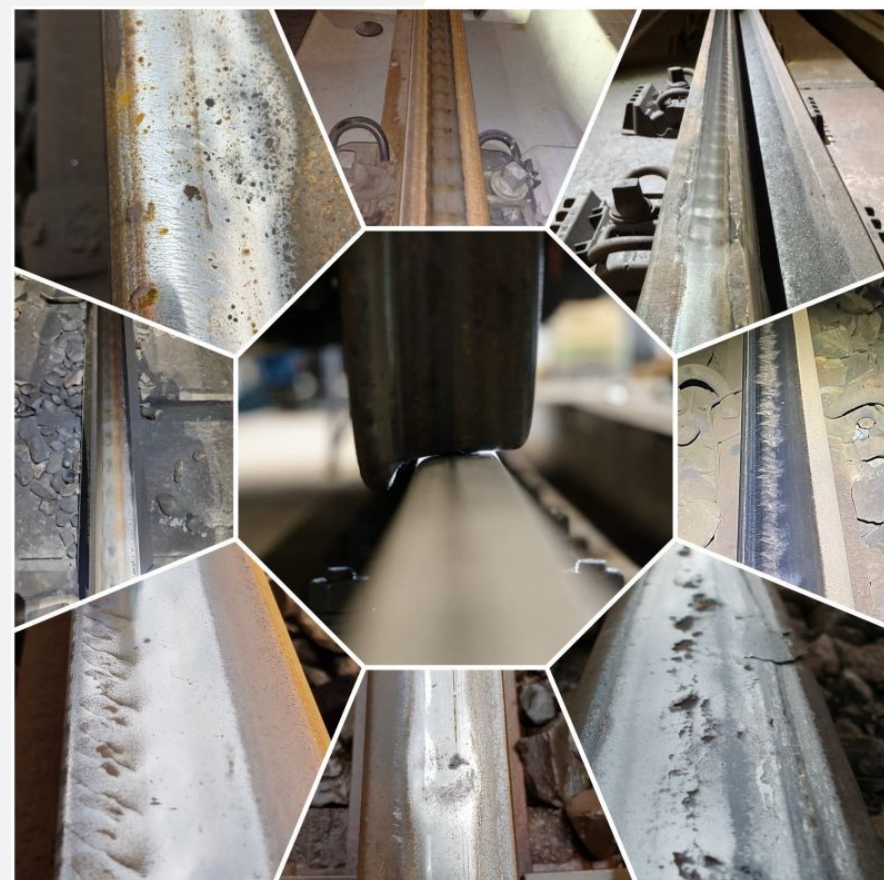
(c.o.f. on wheel flange / gauge corner)



CORRUGATION

?

(promoted by high c.o.f.)



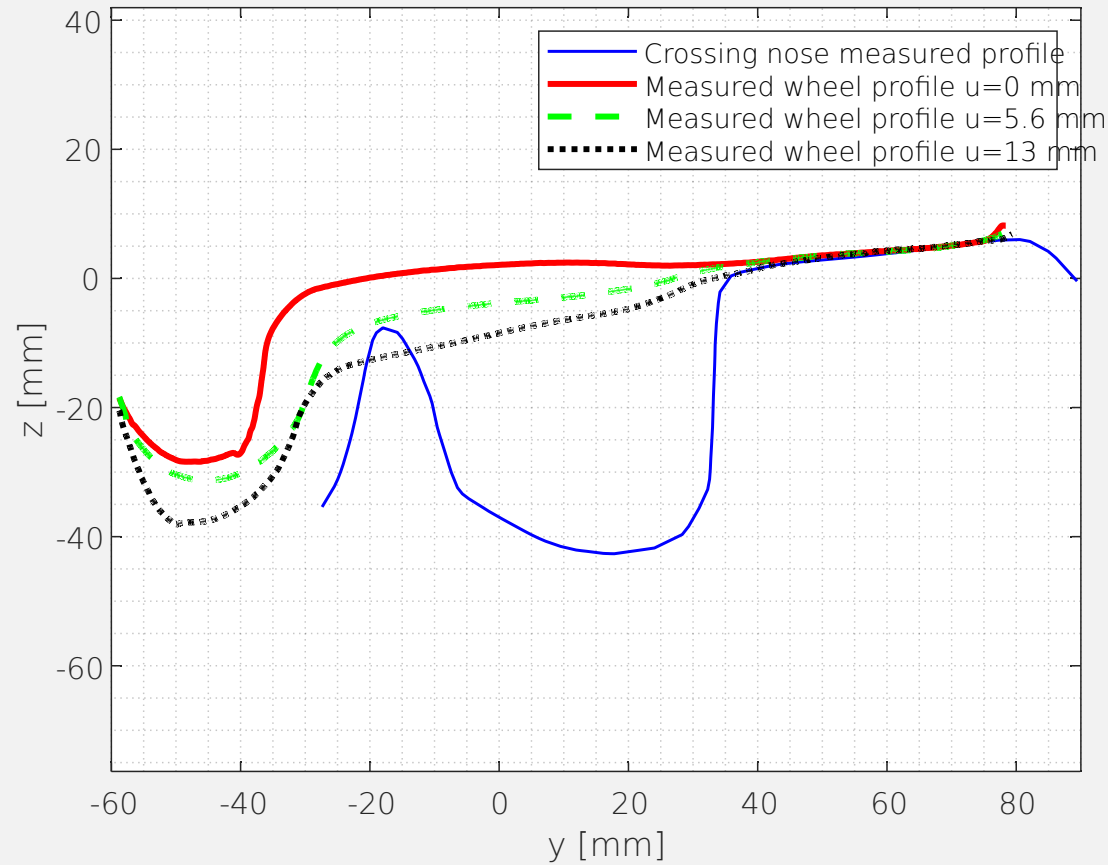
RCF

(high tangential loads)





WHEEL-RAIL ISSUES (or why we are here)



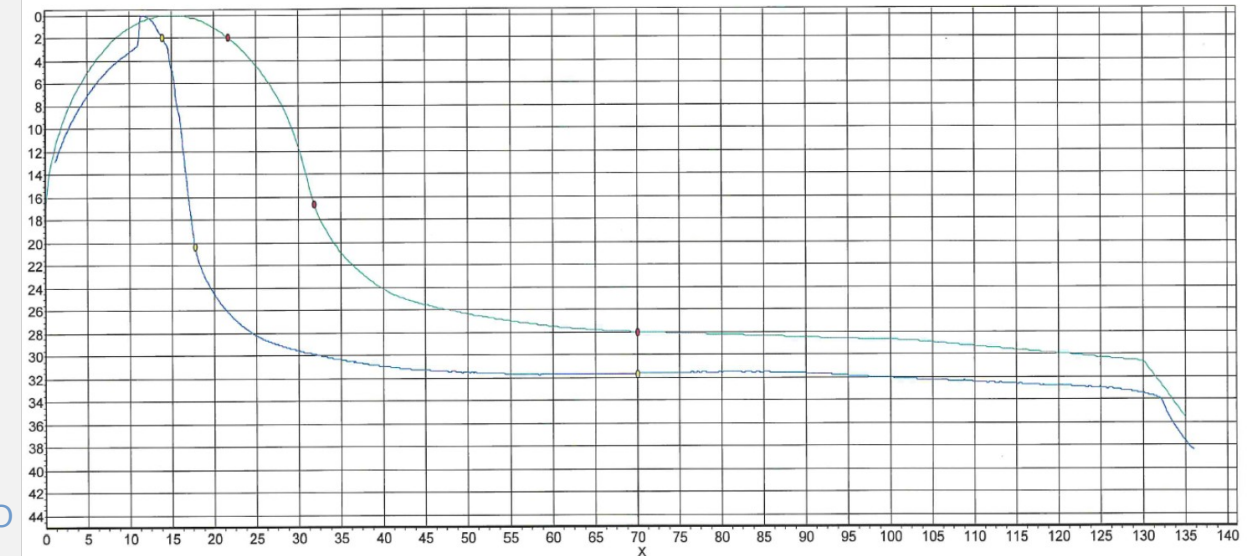
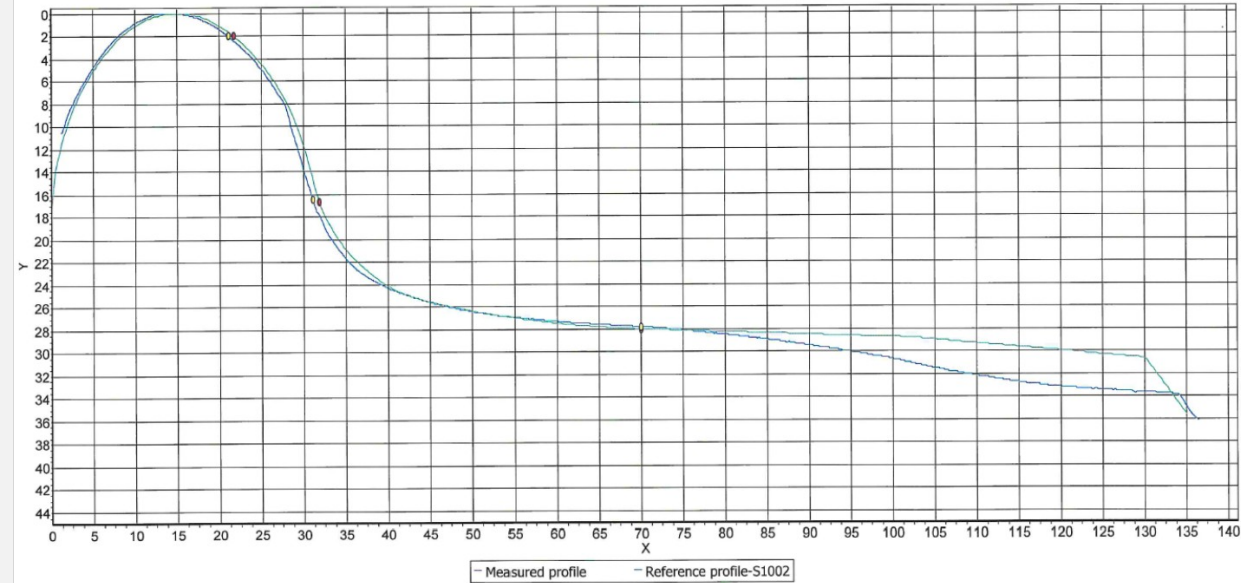
FROGS (POINT) DAMAGE
(wrong wheel profile)



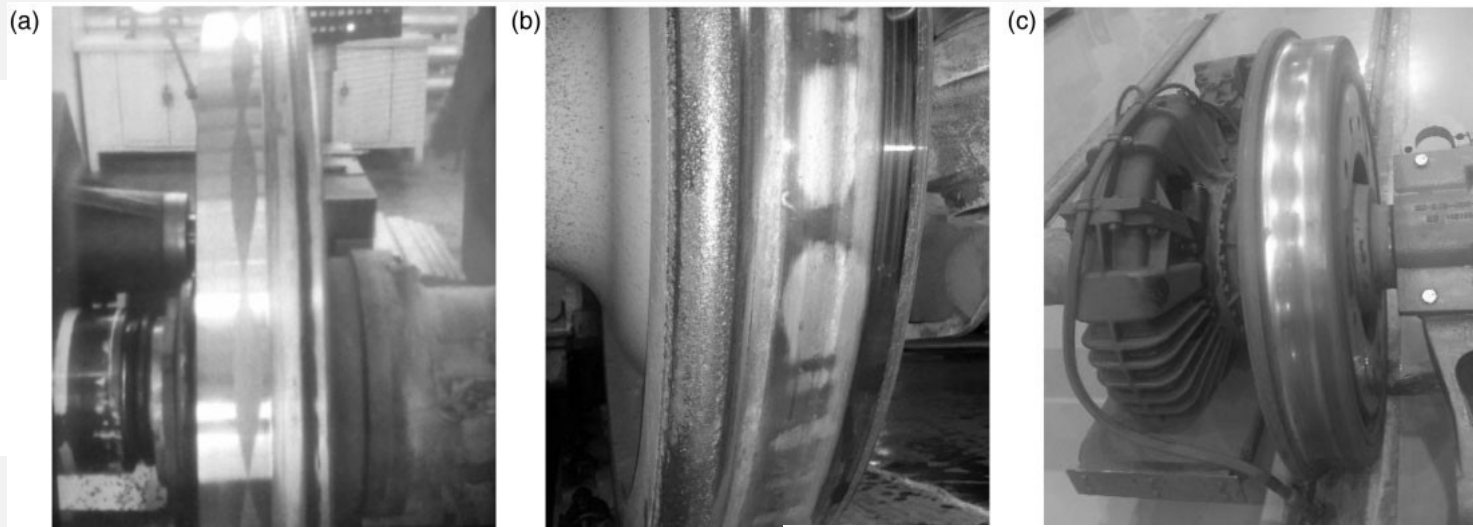
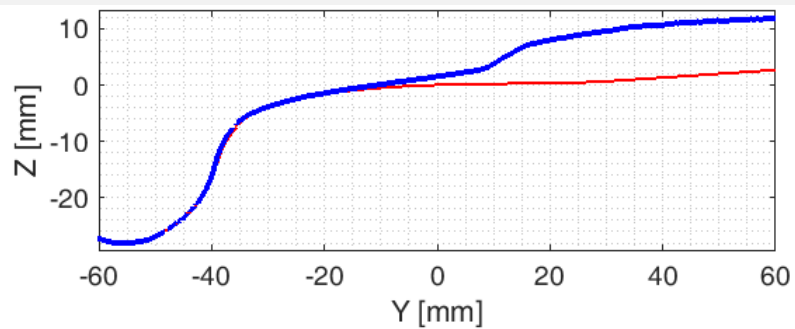
WHEEL-RAIL ISSUES (or why we are here)

FLANGE WEAR

(c.o.f. on wheel flange / gauge corner, asymmetry in the vehicle, wrong a.o.a.)

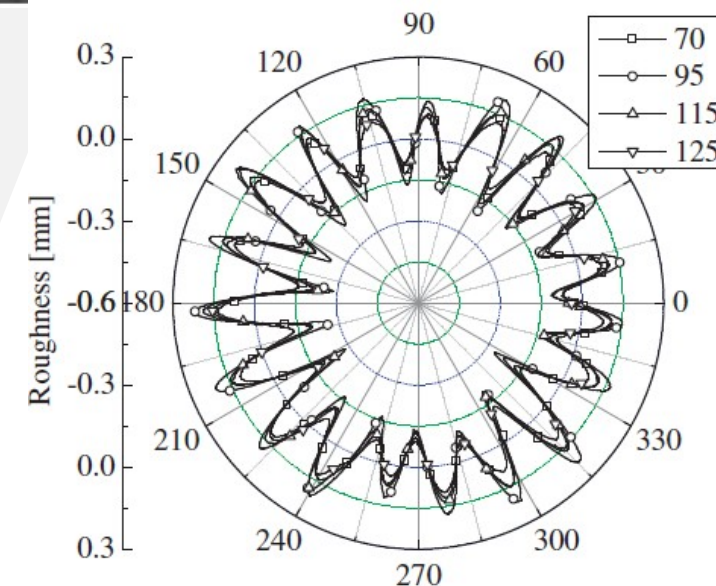


WHEEL-RAIL ISSUES (or why we are here)



WHEEL TREAD WEAR
(wrong gauge, high c.o.f.)

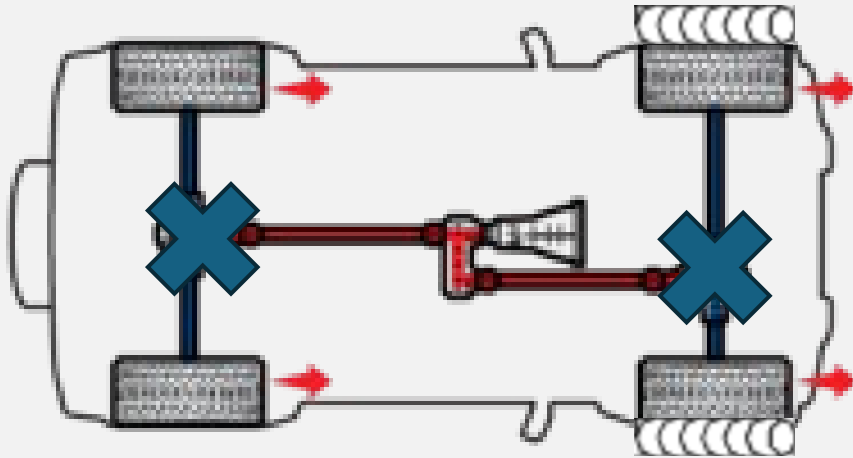
POLYGONIZATION - OOR
(motor – kinematic chain - wheels)



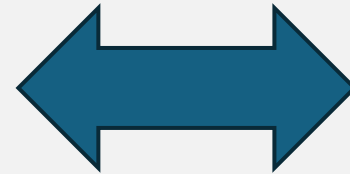


WHEEL-RAIL ISSUES (or why we are here)

AWD car with some
«strange» features



Conventional truck



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RAIL VEHICLE DYNAMICS

4.5 The Conflict Between Steering and Stability

Steering and stability of the bogie: vehicle dynamics and suspension design

A H Wickens

Department of Mechanical Engineering, Loughborough University of Technology



How can we tackle these problems?

- **As in all systems each component may be OK but their interaction dramatically fails**
- **Quite often there is a strong debate between the track and rolling stock departments**
- **Managers rarely have a clear view of what's happening (disappointing...)**
- **ONLY when service regularity is endangered then external expertise is looked for**
- **Someone thinks that the problems may be solved by “the expert” (but nobody has a magic wand!)**



What skills should the expert have?

- Is track engineering enough?
- Is vehicle engineering enough?
- Is vehicle-track interaction (running dynamics) enough?
- Is mechanics of materials enough?
- Is friction management enough?
- Is maintenance engineering enough?
- ...
- ...but the biggest problem is not technical!



How to get the best from employees' skills

- In many cases people work for a PTA since decades
- Listening to expert people is fundamental to create the right mood
- Valuable but limited knowledge (“We don’t know what we don’t measure”)
- The recipe is always the same: education



What should the expert do?

- **Be flexible with the contract**
- **Go on the track, ride the vehicles, take some measurements, «sniff the air»**
- **Identify lack of knowledge and of equipment**
- **Learn which resources are available**
- **Define a strategy to finish the job in the most efficient way**
- **Leave a new structure behind, with skilled people and a clear, unequivocal company structure to continue the job**
- **Remain at the company disposal for whatever may be needed**

But... who is “the Expert”?

Who	Skills	Availability	Pros and cons
University	Top	Limited to the scope of the research	P: Highest knowledge of physics C: out of the core business of a university
Supplier	Limited to the scope of supply	Very good (normally)	P: They (may) fix the problem temporarily C: Don't see the whole picture – cannot identify a solution
Laboratory	State-of-the-art	Excellent	P: They may identify the root cause C: Responsibility limited to test
Expert (aka “one man band”)	Mixed	Limited to Average (no subcontracting possible...)	P: May follow the whole process C: Huge knowledge necessary (no improvisation, please!) C: Useless if he/she doesn't provide a «solution»



“I will survive...”

- **Can a solution always be found? Let’s be frank: NO**
- **Vehicles and tracks cannot be changed**
- **Changes require skill, will, competence, time, money and...**
- **... authorization from safety authorities**
- **... people open to accepting new roles, new methods, new balance**
- **It must be clear from the start that all that can be done is the best attempt to survive with the assets and the staff available**



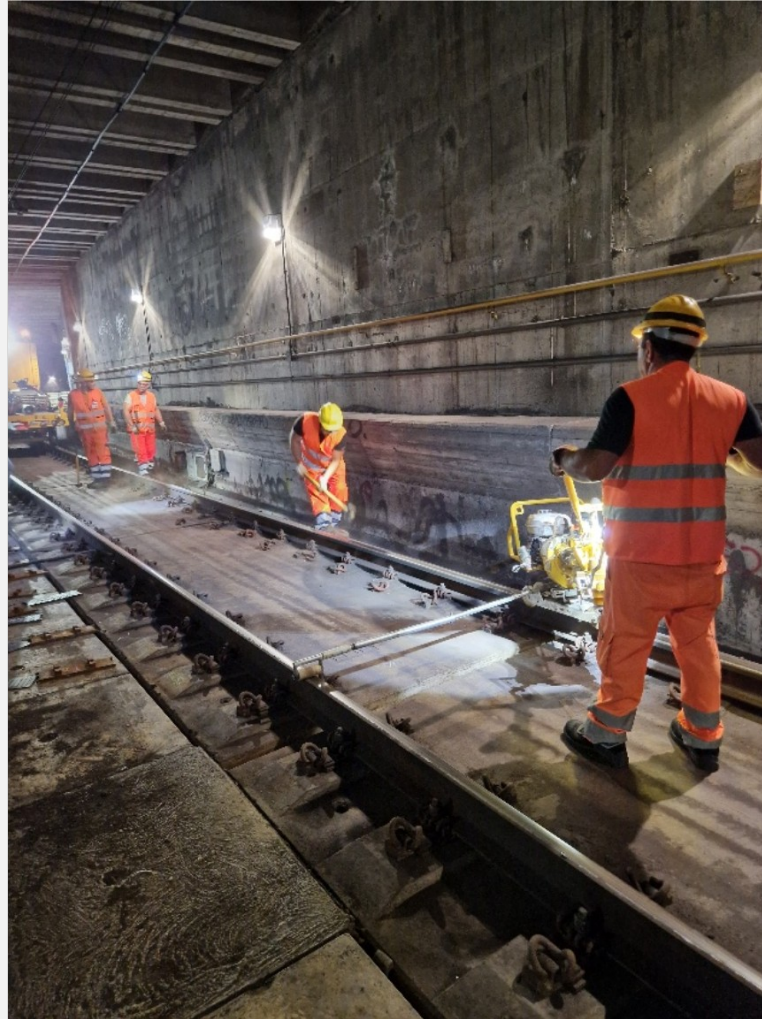
The craziest things I've done in my life

- I have convinced a metro to change the track gauge to improve steering in sharp curves (on a slab track it's not that easy...)
- I removed cleaning brake blocks replacing them with friction modifier sticks (safety case was exceptionally hard, believe me)
- I convinced a metro to run like hell in a curve to reduce wear (sounds crazy, isn't it?)
- We “destroyed” steering properties of a rail profile to improve W/R contact conditions (see below)
- I forced a train supplier to introduce modifications in a vehicle to reduce corrugation in curves (they didn't even know what corrugation is)
- I proposed (still working on this...) to remove HHR and install NHR back to avoid unexpected and extremely serious damage to rails



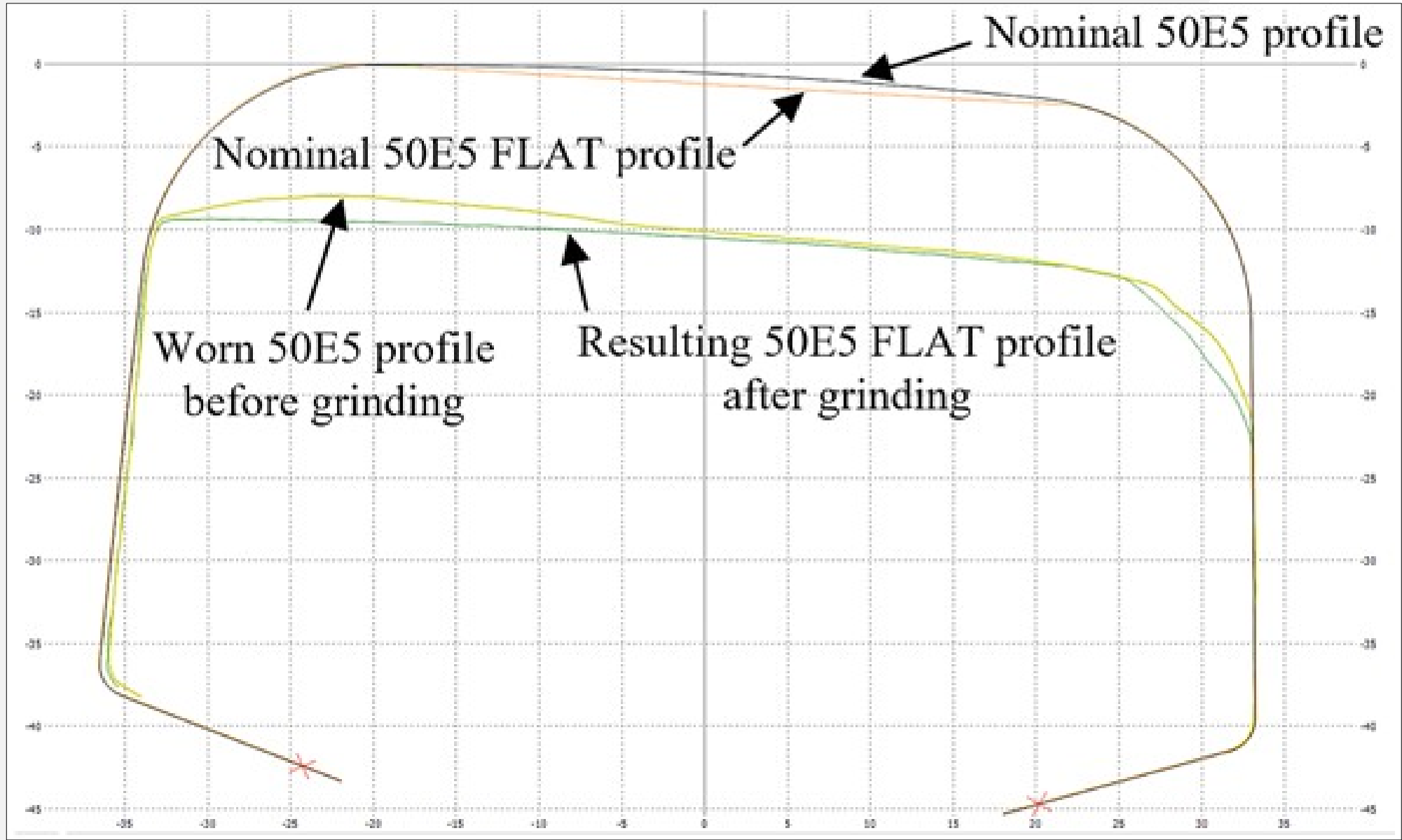


Gauge modification from wide to standard



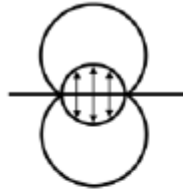


When restoring makes no sense... and theory even less





Happiness is... papers rejected by authoritative journals!



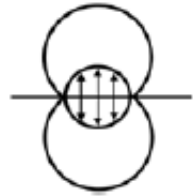
*13th International Conference on Contact Mechanics and Wear of Rail/Wheel Systems
(CM2025), Tokyo, Japan, September 22-26, 2025*

MANAGEMENT OF WHEEL-RAIL DAMAGE IN A TIGHT METRO CURVE

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¹ Dipartimento di Ingegneria Civile e Industriale, Università di Pisa, Italy

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*13th International Conference on Contact Mechanics and Wear of Rail/Wheel Systems
(CM2025), Tokyo, Japan, September 22-26, 2025*

“TURTLE SHELLING”, A NEW TYPE OF DEFECT OF HEAD-HARDENED LOW RAILS IN TIGHT CURVES

A. Bracciali^{*}

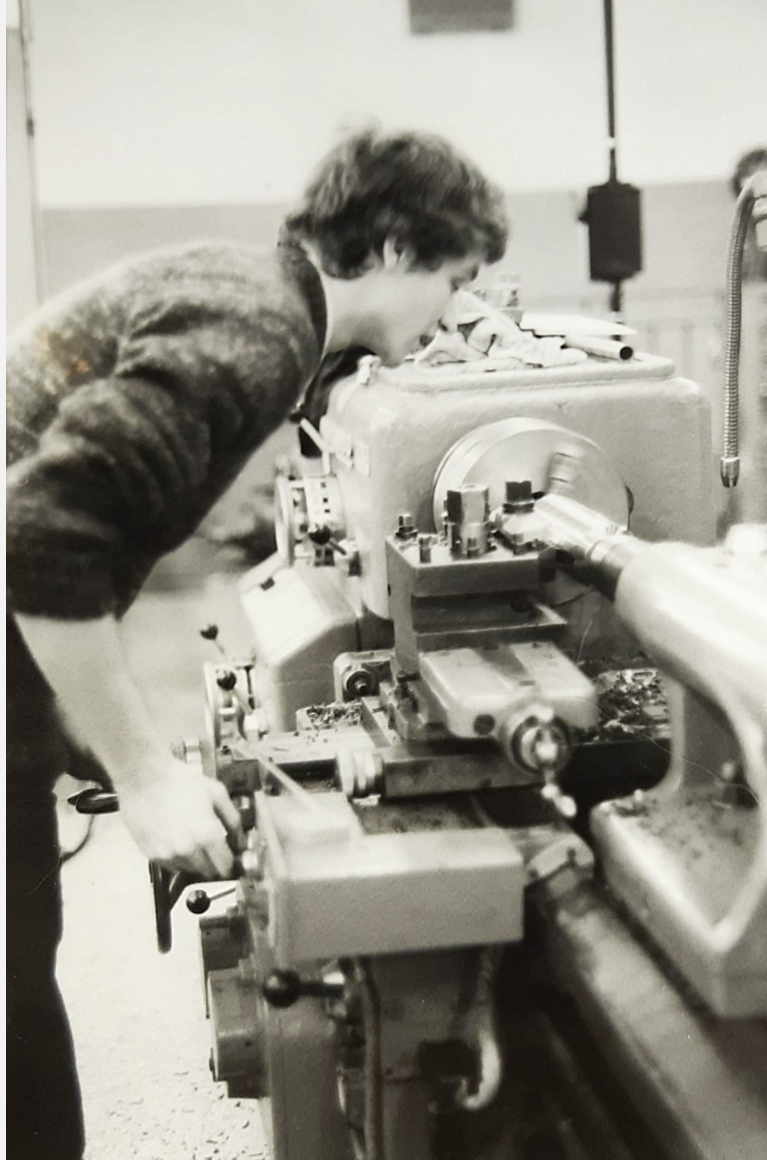
Dipartimento di Ingegneria Industriale, Università degli Studi di Firenze, Italy





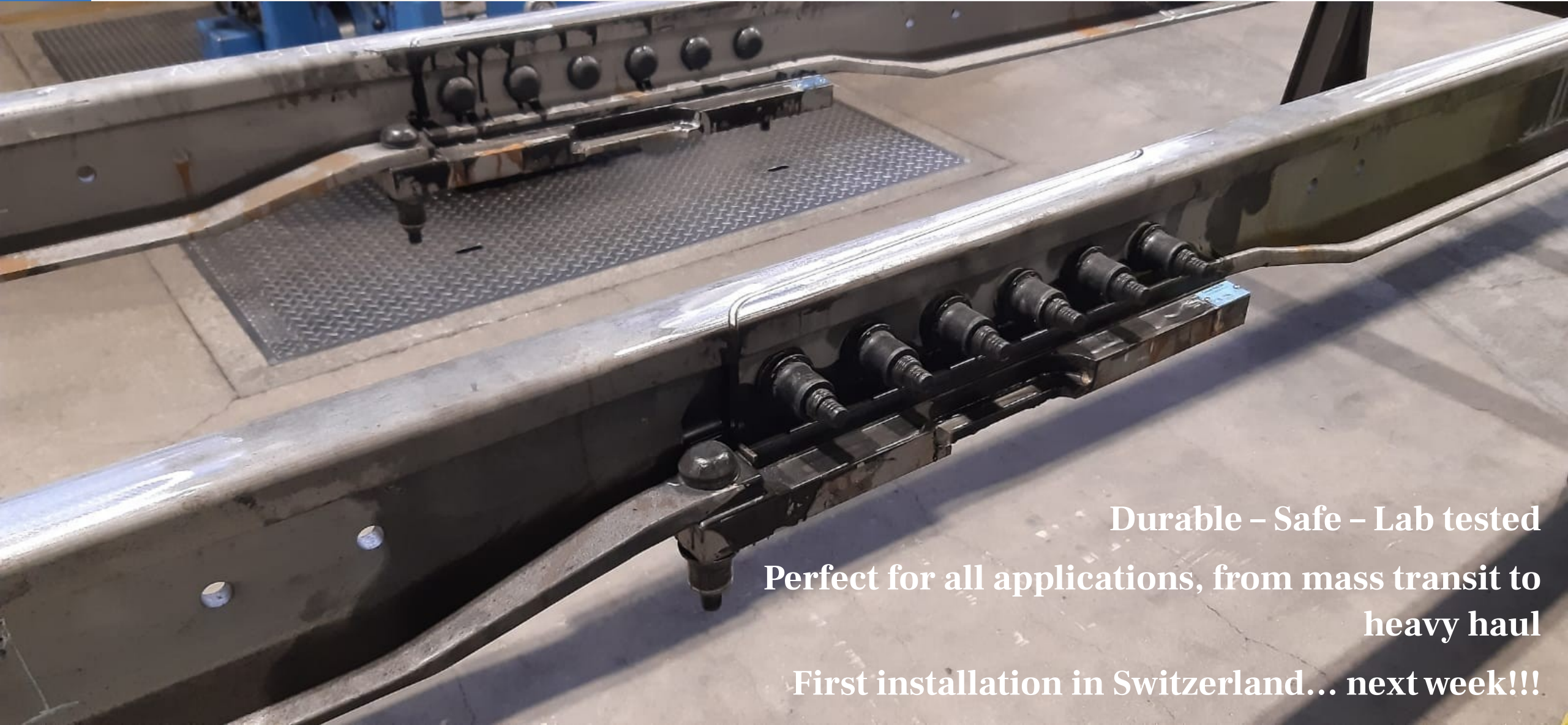
Summarizing...

- **Workshop is where I am from**
- **Theory makes sense only when applied to practical things**
- **After 44 years I still have to learn a lot**
- **Happy to discuss and improve together**



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"Whoever tells you that he has no problems with IRJs is lying" (J. Reinhardt, DB)



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heavy haul
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**The view from my
window at home
and in Boston (spot
the differences)**



RAIL TRA



RAIL TRAN

A photograph of two men in a tunnel. The man on the left is wearing a blue jumpsuit and holding a long metal rod. The man on the right is wearing a dark jacket, a cap, and gloves, also holding a long metal rod. They are standing on a set of railroad tracks. In the foreground, a laptop is placed on a small stand on the tracks, displaying a blue screen with some data. The tunnel has a curved concrete wall and is lit by several bright lights. A red circular sign with a white horizontal bar is visible on the right side of the tunnel.

Thanks for your
attention...
and see you in
North America!